Mitigating Digital E-Waste: Evaluating Software Obsolescence and Frugal Alternatives

Oïlhan Mercul Raphaël Letourneur

October 2026

The environmental footprint of digital technologies is heavily dominated by hardware manufacturing. Premature device replacement is frequently driven by software friction—such as bloatware, heavy updates, and interface slowness—rather than complete physical failure. This project adopts a qualitative approach to examine how legacy devices support common user tasks, evaluate software-based alternatives to extend hardware usability, and assess the broader sustainability benefits of mitigating e-waste.

1 Introduction and Problem Statement

End-user digital devices carry significant embedded environmental costs, largely stemming from raw material extraction and production. While the underlying hardware often remains functional, users frequently discard devices due to software limitations and perceived obsolescence. This project investigates the qualitative capacity of aging hardware to maintain utility, explores software-driven strategies to prolong device lifespans, and evaluates the social and environmental sustainability outcomes of extending technology use.

2 Related Work

Software-induced obsolescence is an incremental process where technical frictions accumulate over time. Mosesso et al. [1] conceptualize this trajectory as an *obsolescence path*, demonstrating that system slowdowns, storage saturation, and update incompatibilities systematically push users toward premature renewal. Furthermore, research on unplanned obsolescence [2] highlights how software maintainability and open ecosystems are critical to extending hardware utility. Building on these insights, our work explores how frugal and alternative software approaches can interrupt these obsolescence paths.

3 Methodology

To systematically address device longevity, this project relies on a qualitative evaluation methodology. Rather than focusing solely on low-level resource benchmarks, we evaluate user-centric task viability, functional completeness, and software responsiveness on legacy hardware.

Task	Description
Identify the main task	Which are the functionalities we want when using one device (computer, phone)
Test the main software	For each functionalities, chose the most famous and used software. Build a testing framework for each
Test the alternative software	For each functionalities, chose the alternative and more frugal software. Build a testing framework for each
Try it	Putting it into practice our testing framework on devices
Synthesize	Build a matrix of our results, try to analyses it, make conclusions
Research work	What are the benefits of usability extension, the purpose of all our work.

4 Milestones

- 20.10.2026 : Project Proposal
- 27.10.2026 : Find what software limitations make them change their devices
- 03.11.2026 : Find what tasks/functionalities people do on their devices.
- 10.11.2026 : Build testing framework for testing the most used software on the specific task identified.
- 17.11.2026 : Find alternative software solution and build their testing framework.
- 24.11.2026 : Start testing on the devices.
- 01.12.2026 : Continue testing.
- 08.12.2026 : Summarize the information and the benchmarks
- 15.12.2026 : Build the matrix
- 22.12.2026 : Research Benefits of usability extension
- 05.01.2027 : Poster and presentation

5 Evaluation Plan

The project will be evaluated based on:

1. The clarity and relevance of the core tasks identified for testing real-world device usability.
2. The rigor of the qualitative assessment criteria used to evaluate software frugality and user experience on aging hardware.
3. The depth of the qualitative synthesis comparing mainstream software against lightweight alternative solutions.
4. The critical reflection on environmental benefits and alignment with the United Nations Sustainable Development Goals (SDG 12: Responsible Consumption and Production, SDG 10: Reduce inequality).

6 Glossary

Qualitative Testing Framework: A practical evaluation protocol assessing whether essential software tasks can be executed acceptably on aging hardware without severe usability bottlenecks.

7 Resources

Primary literature resources include *Obsolescence Paths: living with aging devices* [1] and *Unplanned obsolescence: Hardware and software after collapse* [2]. Legacy hardware devices and open-source software repositories will be utilized throughout the project.

References

[1] L. Mosesso, N. Maudet, E. Nano, T. Thibault, and A. Tabard, "Obsolescence Paths: living with aging devices," in *Proc. Int. Conf. ICT for Sustainability (ICT4S)*, Rennes, France, 2023, pp. 11–21.

[2] *Unplanned obsolescence: Hardware and software after collapse*, Academic Reference / Literature Resource, 2023.